

About me

My current work focuses on understanding and improving the reasoning capabilities of neural networks. More specifically, I study how they can learn to perform tasks for which efficient classical algorithms already exist, and how correct solutions can emerge through training. Understanding these characteristics is crucial for designing models that exhibit a more consistent behavior. In addition to this, I spent more than two years applying machine learning to real-world applications.

Education

Ph.D. in Computer Science University of Waterloo. Advised by Kimion Fountoulakis	2022 – Present GPA: 92/100
M.Sc in Artificial Intelligence and Robotics Sapienza University of Rome	2019 – 2022 GPA: 98/100 (108/110)
B.Sc in Mechanical Engineering Federal University of Santa Catarina	2013 – 2019 Overall course ranking: 92 nd percentile
Exchange Program RWTH Aachen University	2017 – 2018

Research Experience

Amazon <i>Applied Science Intern</i> Hosted by Youxin Zhang and Ruijun Ma	Jun 2025 – Sep 2025 <i>New York, US</i>
Huawei <i>Research Intern</i> Hosted by Guojun Zhang and Yingxue Zhang	Feb 2022 – Jan 2023 <i>Toronto, Canada</i>
Micromed/CERTI Foundation <i>Research Consultant</i>	Feb 2021 – Jan 2022 <i>Florianopolis, Brazil</i>

Publications

[NeurIPS 2025] **Back de Luca, A.**, Giapitzakis, G., & Fountoulakis, K. “Learning to Add, Multiply, and Execute Algorithmic Instructions Exactly with Neural Networks .”
<https://arxiv.org/abs/2502.16763>

[HiDL @ ICML 2025] Giapitzakis, G., **Back de Luca, A.**, & Fountoulakis, K. “Exact Learning of Permutations for Nonzero Binary Inputs with Logarithmic Training Size and Quadratic Ensemble Complexity.”
<https://openreview.net/pdf?id=4RLpgRyJIm>

[ICML 2025] **Back de Luca, A.**, Giapitzakis, G., Yang, S., Veličković, P., & Fountoulakis, K. “Positional Attention: Expressivity and Learnability of Algorithmic Computation.”
<https://arxiv.org/abs/2410.01686>

[ICML 2024] **Back de Luca, A.** & Fountoulakis, K. “Simulation of Graph Algorithms with Looped Transformers”
<https://arxiv.org/abs/2402.01107>

[ICLR 2024] **Back de Luca, A.**, Fountoulakis, K. & Yang, S. “Local Graph Clustering with Noisy Labels”
<https://arxiv.org/abs/2310.08031>

Awards

PGS President’s Graduate Scholarship	2025
OGS Ontario Graduate Scholarship	2025
DGSA Mathematics Domestic Graduate Student Award	2023
M-IMAE Mathematics International Master’s Award of Excellence Scholarship	2022

Teaching

Teaching Assistant at the University of Waterloo	
CS 479/679 – Neural Networks	2025
CS 348 – Introduction to Database Systems	2024
CS 475/675 – Computational Linear Algebra	2024
CS 338 – Computer Applications in Business: Databases	2024
CS 245 – Logic and Computation	2022-2023

Other Professional Experience

NEO Empresarial <i>Engineering Intern</i>	May 2015 – Jul 2018
CERTI Foundation <i>Data Analytics Intern</i>	Summers 2017 and 2018
Fraunhofer Institute for Production Technology <i>Research Intern</i>	Apr 2017 – Jan 2018
Whirlpool – Embraco <i>Procurement Intern</i>	Summer 2016
Numerical Simulation Lab. in Fluid Mechanics and Heat Transfer <i>Research Intern</i>	Feb 2015 – May 2015